Ruideng Zhong

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EDUCATION

Duke University - Durham, NC **Aug 2025 - Present**

- Major in **Electrical and Computer Engineering**, Minor in **Economics**
- Cumulative GPA: 4.0/4.0; Major GPA: 4.0/4.0

WORK EXPERIENCE

Electrical Systems Engineer - Duke Motorsports Team (FSAE) **Aug 2025 - Present**

- Designed and fabricated vehicle and dynamometer wiring harnesses using RapidHarness.
- Validated continuity, sensor data, and safety requirements via MoTeC C185 Dash Manager and M1 Tune.
- Used Altium to design and verify PCBs for the steering wheel, BSPD, strain gauge, and light gates.

Quality Engineering Intern - EEEA Inc. **May 2026 - Aug 2026**

- Achieved 98% accuracy in classifying assembled PCBs according to IPC-A-610 criteria.
- Led procedure documentation efforts in a push to achieve AS9100 certifications.
- Reduced AOI false-positive rate by 24% through custom programming and setup.

Lab Aide - Clemson University Department of Electrical and Computer Engineering **Aug 2023 - Aug 2025**

- Developed and optimized 3D printing workflows for dentures and biological scaffolds.
- Optimized material selection to enhance the durability of the 3D-printed denture model.
- Conducted a literature review on filament properties for scaffold fabrication and organized data into a database.

Research Intern - Clemson University Department of Mechanical Engineering **Aug 2023 - Aug 2024**

- Adopted molecular dynamics simulations to investigate the material behavior of C_{60} clusters under compression.
- Optimized C++ programs to reduce runtime by 17% using Clemson's Palmetto 2 HPC cluster.
- Analyzed outputs and logs to validate results and identify anomalies using Python.

PROJECTS

Custom Hytera Radio Harness Using IMSA Connector **Apr 2026 - Present**

- Designing and fabricating a custom vehicle wiring harness to interface a Hytera radio with an M1 connector, recreating the functionality of commercial motorsports radio harnesses.
- Researched compatible components and wires to develop a wiring schematic and pinout using KiCAD.

Gate-Level 16-Bit MIPS-Style CPU **Feb 2026 - Apr 2026**

- Implemented a 16-bit CPU in Logisim with a custom ALU and a multicycle control unit.
- Designed gate-level registers, counters, and shifters with component constraints.
- Supported arithmetic, logical, memory, and control-flow instructions.

Hardware Acceleration with Verilog on AMD FPGA Platform **June 2024 - Aug 2025**

- Designed and implemented digital circuit interfaces in Verilog to accelerate targeted computing workloads.
- Deployed designs on an AMD FPGA development board, achieving an 8% performance improvement.

SKILLS

- **Software:** Altium, Cadence OrCAD, KiCAD, Logisim, RapidHarness, MoTeC M1 Tune, SolidWorks
- **Programming and Systems:** Python, Java, C, C++, MATLAB, Verilog, Git, Linux, Unix
- **Laboratory Equipment:** Digital Multimeters, Oscilloscopes, Function Generators, Soldering
- **Productivity:** Microsoft 365 (Excel, PowerPoint, Word), Google Suite, Airtable, Trello

ADDITIONAL INFORMATION

- **Relevant Coursework**: Computer Architecture, Microelectronics, Signals and Systems, Data Structures and Algorithms, Differential Equations, Linear Algebra, Multivariable Calculus, Electricity and Magnetism Physics
- **Awards:** "Most Innovative Design" Award, Pratt Design Expo (2025)
- **Activities:** Duke Marching and Pep Bands (Treasurer), ECE 250 Teaching Assistant (Computer Architecture)